INCH-POUND
MIL-PRF-49080C(CR)
26 March 2007
SUPERSEDING
MIL-P-49080B(CR)
8 February 2002

# PERFORMANCE SPECIFICATION

# POWER SUPPLIES, PP-2953D/U AND PP-6224C/U

This specification is approved for use by the Communications-Electronics Life Cycle Management Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1.0 SCOPE

1.1 Scope. This specification establishes the performance and verification requirements for each of the following Power Supplies: the PP-2953D/U and the PP-6224C/U . The power supply furnishes 0-25 amperes over a voltage range of 24 to 32Vdc from nominal 115/230 Vac at 50, 60 or 400 Hz single phase. The power supply can also operate from a 24 Vdc storage battery in the battery standby mode.

## 2.0 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4 or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army CE-LCMC, ATTN: AMSEL-LC-LEO-E-AV, Fort Monmouth, New Jersey 07703-5023. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SPECIFICATIONS

DEPARTMENT OF DEFENSE

| | | |
|---|---|---|
| MIL-PRF-31032 | - | Printed Circuit Board/Printed Wiring Board, General Specification For |
| MIL-C-53039 | - | Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant |
| MIL-C-53072 | - | Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection |
| MIL-DTL-64159 | - | Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant |

STANDARDS

FEDERAL

| | | |
|---|---|---|
| FED-STD-595/34094- | | Colors Used in Government Procurement |

DEPARTMENT OF DEFENSE

| | | |
|---|---|---|
| MIL-STD-461 | - | Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment |
| MIL-STD-810 | - | Environmental Engineering Considerations and Laboratory Tests |
| MIL-STD-3009 | - | Lighting, Aircraft, Night Vision Imaging System (NVIS) Compatible |
| MIL-STD-13231 | - | Marking of Electronic Items |

(Copies of these documents are available online at http://assists.daps.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).



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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

| | | |
|---|---|---|
| A3286985 | - | PP-2953D/U & PP-6224C/U Housing Assembly/Mounting Requirements |
| SM-D-667917 | - | Cable Assembly, Power, Electrical CX-11979/U |
| SM-B-667921 | - | Connector, Receptacle |
| SM-D-667953 | - | Cable Assembly, Power, Electrical CX-12342/U |

(Copies of this drawing are available from the US Army Communications-Electronics Command, ATTN: AMSRD-CER-C2-AP-PA, Fort Monmouth, NJ 07703.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subject to first article inspection in accordance with 4.2.

3.2 Parts and Materials. The contractor shall select the materials, but the materials and parts used shall be fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material shall be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard.

3.3 Design/Construction. The equipment shall be designed to obtain minimum size and weight with maximum reliability consistent with the overall requirements. The connectors and cables for the power supply shall be as specified in this specification. Design techniques shall be used that provide stability under static and dynamic load conditions and minimize Electromagnetic Interference (EMI). The functional requirements of the Power Supply shall be as specified in this specification.

3.3.1 System Configuration. The power supplies shall include the following cable assemblies as part of the system configuration:

| Power Supply | AC Input Cable/Qty (see para 3.3.3.1) | DC Output Cable/Qty (see para. 3.3.3.2) | Battery Standby Cable/Qty (see para. 3.3.3.3) |
|---|---|---|---|
| PP-6224C/U | 1 per system | 1 per system | 1 per system |
| PP-2953D/U | 1 per system | 2 per system | 1 per system |



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3.3.2 Dimensions. The envelope dimensions (chassis) of the Power Supply shall conform to Drawing A3286985 to ensure the mounting requirements (for bench-mounted capability) and the form and fit of the device. The physical dimensions of the Power Supply shall not exceed the following:

| | |
|---|---|
| Height | 7.0 inches |
| Width | 14.25 inches |
| Depth | 14.50 inches |

3.3.2.1 Cover. A cover shall be installed to protect the front panel.

3.3.3 Cables.

3.3.3.1 AC Cable. The AC Input Cable shall be the CX-11979( )/U. The CX-11979( )/U shall conform to the dimensional requirements of Drawing SM-D-667917 to ensure proper interface requirements. This drawing is referenced for dimensional and interface purposes only.

3.3.3.2 DC Cable. The DC Output Cable shall be the CX-12342( )/U. The CX-11979( )/U shall conform to the dimensional requirements of Drawing SM-D-667953 to ensure proper interface requirements. This drawing is referenced for dimensional and interface purposes only.

3.3.3.3 Battery Standby Cable. The contractor shall fabricate the cable used to power the 24 Vdc storage battery in the standby mode (see 3.6.1.1 and 3.6.1.2). This cable shall be capable of mating with the dc input connector on the power supply ( see 3.3.4.1).

3.3.4 Connectors.

3.3.4.1 DC Input Connector. A connector (Part Number MW20 M(M)A00 or equivalent) shall be provided for DC input in the battery standby mode ( see 3.6.1.1 and 3.6.1.2) . This connector shall be located on the rear panel of the power supply.

3.3.4.2 AC Input Connector. One AC input plug connector shall be provided on the front panel. The AC input connector (Part Number PT07C14-5P or equivalent) shall be capable of mating with the AC Cable, CX-11979( )/U.

3.3.4.3 DC Output Connector. Two DC Output Connectors (Part Number MW20 F(M)A00 or equivalent) shall be provided for dc output power with remote sensing capability. Both connectors shall be located on the rear panel of the power supply.

3.3.4.4 DC Output Connector (Front Panel). One DC Output Connector shall be provided in accordance with the dimensional requirements of Drawing SM-B-667921. This connector shall be located on the front panel of the power supply. Drawing SM-B-667921 is referenced for dimensional and interface purposes only.

3.3.5 Indicator Lights/Output Meters. All indicator lights and output meters used shall be Class B Night Vision Imaging System (NVIS) (see 6.5.7) compatible in accordance with the requirements of MIL-STD-3009 for “warnings, caution, and advisory displays and indicators” used on ground vehicles and equipment (see paragraphs 3.1.4, 3.2.2 (secure), 3.2.3 (friendly), 3.2.4 (covert), 3.6, and Appendix C of MIL-STD-3009). NVIS compatibility assures the operation of the power supply without degrading the image intensification capabilities of the NVIS during night operations. Indicator lights and output meters used shall also maintain daylight readability and legibility.

3.3.5.1 DC Emergency Indicator Light. The DC Emergency Indicator Light shall indicate that the power supply is operating in the Battery Standby Mode (see 3.6.1.1 and 3.6.1.2).



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3.3.5.2 Normal Indicator Light. The Normal Indicator Light shall indicate the power supply is producing regulated dc output using ac input.

3.3.5.3 AC Indicator Light. The AC Indicator Light shall indicate that ac input is being provided to the power supply.

3.3.5.4 Output Meter. The Output Meter shall be capable of displaying dc output voltage or current.

3.3.5.4.1 Output Meter Switch. The Output Meter switch shall provide the capability to select either the output voltage or output current to be displayed on the output meter.

3.3.6 DC Adjustment. When operating in the AC input mode, the DC Adjustment shall allow for desired dc output voltage.

3.3.7 115V-230V Switch. The switch shall provide the capability to select the appropriate ac input voltage.

3.4 Weight. The total weight of the Power Supply shall not exceed 36 pounds (excluding the cables). The total weight of the Power Supply in its packaging and shipping requirements shall not exceed 44 pounds.

3.5 Printed Wiring Boards and Printed Wiring Assemblies. Printed wiring boards and printed wiring assemblies used shall conform to the requirements of MIL-PRF-31032.

3.5.1 Repair authorization. Except upon prior specific approval of the contracting officer, printed wiring assemblies shall not be repaired.

3.5.2 Printed wiring boards/assemblies. Printed wiring boards shall be QML certified as specified in MIL-PRF-31032.

3.5.3 Marking. All printed circuit boards shall be marked per MIL-STD-13231 with the following information: Manufacturer's part number, CAGE CODE, and NSN (if applicable).

3.6 Performance Characteristics. The Power Supply shall be capable of providing power for up to two loads and shall meet the following electrical requirements:

3.6.1 Input Voltage (AC). The power supply shall deliver the output required by this specification (see 3.6.2) from the following single phase input sources:

| Volts ac (± 10%) | Frequency –Hz (±5%) |
|---|---|
| 115/ 230 | 50, 60 and 400 |

The power supply shall also operate from a 24 Vdc storage battery (standby operation, see 3.6.1.1 and 3.6.1.2).

3.6.1 1 Battery Standby Mode (115 Vac). The power supply shall automatically switch to the Battery Standby Mode of operation when the 115 Vac input is lost or drops below 77 Vac (plus 12, minus 0 Vac) . This mode produces unregulated dc output. This mode of operation shall be indicated by an indicator light on the front panel labeled "DC EMER" and extinguishing of the "DC NORM" indicator light. When the ac recovers to 103 Vac (plus 0, minus 12 Vac) , or greater, the power supply shall automatically switch to ac operation with controlled dc output. The DC Norm Indicator Light shall turn on. The DC Emergency Indicator Light shall turn off. (see 3.6.5.3)



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3.6.1.2 Battery Standby Mode (230 Vac). The power supply shall automatically switch to the Battery Standby Mode of operation when the 230 Vac input is lost or drops below 154 Vac (plus 24, minus 0 Vac). This mode produces unregulated dc output. This mode of operation shall be indicated by a light on the front panel labeled “DC EMER” and extinguishing of the “DC NORM” indicator light. When the ac recovers to 206 Vac (plus 0, minus 24 Vac), the power supply shall automatically switch to ac operation with controlled dc output. The DC Norm Indicator Light shall turn on. The DC Emergency Indicator Light shall turn off. ( see 3.6.5.3)

3.6.2 Output Voltage. The voltage across the output of the power supply shall be 28.0 Volts (±1%) over the load range of 0-25 amperes. The output voltage shall also be adjustable over the range of 24.0 to 32.0 volts by an external locking type control and shall be within ±1% of the voltage setting over the load range from 0 to 25 amperes (see 4.5.1).

3.6.3 Ripple voltage. The rms ripple voltage shall not exceed 0.5 percent at any output condition from no-load to full-load. The peak to peak ripple voltage including switching spikes and other noise components shall not exceed ±1% at any output condition from no-load to full-load. The ripple voltage includes voltage perturbations within the frequency range of 20 Hz to 10 MHz. ( see 4.5.2)

3.6.4 Recovery Time. The output voltage of the power supply shall recover to 28.0 Volts ±1% within 25 milliseconds when the load is switched from no-load to full-load and from full-load to no-load. The maximum instantaneous overshoot voltage under these conditions and for any turn on or turn off condition of the power supply shall not exceed 20% of the nominal voltage (28.0 Vdc). (see 4.5.3).

3.6.5 Protection.

3.6.5.1 General. The power flow conversion circuitry and feedback control circuitry of the power supply shall be fully protected under operational and abnormal conditions which include the following:

Short circuit of dc output.
Reverse polarity of the battery at the output.
Progressive and prolonged output overloads.
Over voltage line surges and transient voltage spikes.
Under voltage of the input voltage source below the minimum specified.

The equipment shall automatically resume normal operation when any of the above faults are removed (see 4.5.6 and 4.5.11).

3.6.5.2 Overload protection. The power supply shall be protected from physical and electrical damage from any overload conditions (including a direct short circuit) by the use of re-settable circuit breakers in the “ac input” circuit and the “dc output” circuit, and by automatic current limiting circuitry. (see 4.5.4).

3.6.5.3 Power Changeover. The power supply shall include an under voltage relay which shall automatically transfer the “OUTPUT” connector to receive power from the “DC INPUT” connector when the 115Vac input is lost or varies to a low of 77 Vac (plus 12, minus 0 Vac), or the 230 Vac input is lost or varies to a low of 154 Vac (plus 24, minus 0 Vac). (see 3.6.1.1 and 3.6.1.2). The dc power supply shall continue to deliver power at an output voltage of no less than 19 volts at the minimum input changeover voltage. The relay shall return the “OUTPUT” connector circuit to normal (AC Input Mode) when the 115 Vac input is restored or rises to 103 Vac (plus 0, minus 12 Vac), or if the 230 Vac input is restored or rises to 206 Vac (plus 0, minus 24 Vac). (see 3.6.1.1 and 3.6.1.2). There shall be consistency of polarity of the circuit connecting the “battery in” plug and the “power out” plug. (see 4.5.5)

3.6.5.4 Current limiting. Automatic control circuitry shall limit the output current to a maximum of 30 amperes to protect the power supply from overload and automatically restore the output when the overload is removed. (see 4.5.4)



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3.6.5.5 Battery discharge. The power supply shall include provisions to prohibit the discharge of the battery used for "DC Emergency power" under the conditions of:

Loss of power input to the charger.
Battery emf in excess of dc output voltage.

3.6.6 Efficiency. The overall efficiency of the power supply shall not be less than 70% (see 4.5.8)

3.6.7 Remote sensing. The power supply shall include provisions for extending the sensing of the dc output voltage through signal leads from the auxiliary output receptacle to the remote load terminals. Also, provisions shall be included to automatically revert to local sensing if the remote circuit is inadvertently opened. (see 4.5.7)

3.6.8 Stability. Output voltage drift at a constant load and ambient temperature shall not exceed 0.5% over a period of 24 hours. (see 4.5.9)

3.6.9 Temperature coefficient. Change in output voltage as a function of ambient temperature shall not exceed 0.02 percent per degree F (0.036 percent per degree C) over the entire specified operating temperature range. (see 4.5.10)

3.7 Environmental conditions. The equipment shall meet the following environmental conditions:

3.7.1 Temperature. The equipment shall operate at ambient temperature in the range of +150°F (+66°C) to -40°F (-40°C). (The 150°F (66°C) temperature includes effect of solar loading.) Exposure during operation at the high temperature extreme shall not exceed 4 hours and at the low temperature extreme shall not exceed 72 hours, at any one time. The equipment (non-operating) shall withstand exposure in the range of +160°F (+71°C) to -80°F( -62.2°C), exposure at the high temperature not to exceed 48 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.7.1.1 Low. Before, during, and after testing as specified in 4.6.1.1, the equipment shall meet full specification performance for the following measurements:

| Measurement | Reqt Para. | Test Para. |
|---|---|---|
| Input and Output Voltage | 3.6.1, 3.6.2 | 4.5.1 |
| Ripple Voltage | 3.6.3 | 4.5.2 |
| Recovery Time | 3.6.4 | 4.5.3 |
| Power Changeover | 3.6.5.3 | 4.5.5 |

3.7.1.2 High. Before, during, and after testing as specified in 4.6.1.2, the equipment shall meet full specification performance for the following measurements:

| Measurement | Reqt Para. | Test Para. |
|---|---|---|
| Input and Output Voltage | 3.6.1, 3.6.2 | 4.5.1 |
| Ripple Voltage | 3.6.3 | 4.5.2 |
| Recovery Time | 3.6.4 | 4.5.3 |
| Power Changeover | 3.6.5.3 | 4.5.5 |



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3.7.2 Humidity. The equipment shall withstand up to 98 percent relative humidity for ten days. When tested as specified in 4.6.2, the Power Supply shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet full specification performance during the specified times for the following measurements:

| Measurement | Reqt Para. | Test Para. |
|---|---|---|
| Input and Output Voltage | 3.6.1, 3.6.2 | 4.5.1 |
| Ripple Voltage | 3.6.3 | 4.5.2 |

3.7.3 Low pressure (Altitude). The equipment shall operate up to 15,000 feet above sea level and shall withstand up to 40,000 feet above sea level. Full specification performance for the following measurements shall be met before, during, and after testing as specified in 4.6.3:

| Measurement | Reqt Para. | Test Para. |
|---|---|---|
| Input and Output Voltage | 3.6.1, 3.6.2 | 4.5.1 |
| Ripple Voltage | 3.6.3 | 4.5.2 |

3.7.4 Immersion. The equipment shall show no evidence of leakage after it is tested as specified in 4.6.4.

3.7.5 Vibration and Ballistic Shock. The equipment shall meet the requirements of Table I.

TABLE I. Vibration and Ballistic Shock.

| Inspection | Inspection Paragraph | Performance after test |
|---|---|---|
| Vibration | 4.6.5 | Specified performance (see Note 1) |
| Shock test; ballistic | 4.6.6 | Specified performance (see Note 1)<br>Any physical damage shall be minor only. |

Note 1: Prior to and after testing, the equipment shall meet the specified performance for the following requirements: 3.6.1, 3.6.2, 3.6.3, and 3.6.5.3.

3.7.6 Fungus. After the fungus test of 4.6.7, the Power Supply shall be visually examined, using a 10 power magnifier. The Power Supply shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.5.3) as evidenced by growth colonization (see 6.5.2) which includes branching ( see 6.5.1) and sporulation (see 6.5.4) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.5.5) shall not be included in this evaluation.

3.7.7 Sand and dust. The equipment shall withstand the following:

Blowing sand. Particle size .01 to 1.0 mm with wind speeds of 15 knots or greater.
Blowing dust. Particle size .0001 to .01 mm with wind speeds of 15 knots or greater.

Before and after testing as specified in 4.6.8, the equipment shall meet full specification performance for the following measurements:

| Measurement | Reqt. Para. | Test Para. |
|---|---|---|
| Input and Output Voltage | 3.6.1, 3.6.2 | 4.5.1 |
| Ripple Voltage | 3.6.3 | 4.5.2 |



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3.8 Electromagnetic Interference (EMI). The equipment shall comply with the following emissions and susceptibility requirements of MIL-STD-461. (see 4.9)

TABLE II. Electromagnetic Interference.

| Requirement | Description | Remarks |
|---|---|---|
| CE101 | Power Leads Conducted Emissions, 30 Hz to 10kHz | |
| CE102 | Power Leads Conducted Emissions, 10kHz to 10 MHz | |
| CS114 | Bulk Cable Injection, Conducted Susceptibility, 10kHz to 200 MHz | |
| RE102 | Electric Field Radiated Emissions, 10kHz to 18GHz | RE102 shall be performed from 14 kHz to 1 GHz |
| RS103 | Electric Field, Radiated Susceptibility, 2MHz to 40GHz | See Note below* |

*Note: RS103 shall be performed from 10 KHz to 400 MHz as follows:

| Frequency Range | Susceptibility Level |
|---|---|
| 10 KHz to 1.99 MHz | 1 Volt/meter |
| 2.0 MHz to 29.99 MHz | 5 Volt/meter |
| 30 MHz to 400 MHz | 10 Volt/meter |

EMI filters shall be required for both the AC input and the DC input.

3.9 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment( see 4.7). Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.10 Air-seal test, provision for. The case shall provide means for enabling performance of the air-seal test(see 4.6.9 and 6.4).

3.11 Reliability. The power supply shall have a 90% probability of providing 5,000 hours of failure-free operation under the following conditions:

Ambient temperature of 122°F (50°C).
Duty cycle-intermittent operation.
Environment-ground mobile.

The above requirement shall be verified in accordance with 4.10.



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3.12 Workmanship. Workmanship shall be in accordance with 3.12.1 for the following:

general requirements for plastic material and plastic parts
mechanical operation of controls
printed wiring assemblies and printed wiring boards
screws & other threaded devices and related parts
cleaning
riveting
soldering
welding
internal wiring practices including: slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable)
Marking
Finish, protective
CARC

3.12.1 Workmanship Requirements. This requirement established the acceptance workmanship criteria for electronic equipment.

3.12.1.1 Cleaning. After fabrication, parts and assembled equipment shall be cleaned of smudges; loose, spattered or excess solder; weld metal; metal chips and mold release agents; or any other foreign material which might detract from the intended operation, function or appearance of the equipment.

3.12.1.2 Threaded part or devices. Screws, nuts and bolts shall show no evidence of cross threading, mutilation, or detrimental or hazardous burrs, and shall be firmly secured.

3.12.1.3 Bearing assemblies. Bearing assemblies shall be free of rust, discoloration, and imperfections of ground, honed, or lapped surfaces. Contacting surfaces shall be free of tools marks, gouge marks, nicks, or other surface-type defects. There shall be no detrimental interference, binding or galling.

3.12.1.4 Wiring. Wires and cables shall be positioned or protected to avoid contact with rough or irregular surfaces and sharp edges and to avoid damage to conductors or adjacent parts.

3.12.1.5 Shielding. Shielding on wires and cables shall be secured in a manner that will prevent it from contacting or shorting exposed current-carrying parts. The ends of the shielding or braid shall be secured to prevent fraying.

3.12.1.6 Exterior Surfaces. All exterior surfaces shall be lusterless.

3.13 Requirements for Applying Chemical Agent Resistant Coating (CARC).

3.13.1 Technical Requirements. Exterior surfaces of the Power Supply shall be painted with Lusterless Green 383 CARC in accordance with MIL-DTL-64159 or MIL-C-53039 (FED-STD-595 Color No. 34094). All coatings shall be obtained from a Qualified Products List (QPL) source. Materials and processes used in the application of all preconditioning, passivation, pretreatments, primers and final films shall be in accordance with MIL-C-53072 and shall meet all Verification Provisions of MIL-C-53072.

3.13.2 Visual Inspection Criteria(Verification). Color match on the finished assembly shall be per paragraph 3.3.3 of MIL-DTL-64159 or MIL-C-53039, as applicable.

3.13.3 Touch Up. When piece parts have been finished with complete paint finishes, fixed assemblies shall be touched up in order to provide a continuous film across the interfaces. This requirement does not apply to hairline joints between matching parts, gasketed joints, or where moving parts such as hinges or catches are present. Touch-up paint used shall be identical to the final paint film and shall provide a reasonable color match with that of other assembled parts.



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3.13.4 Coating Thickness. The dry film thickness of each pretreatment, primer and final film coating shall be in accordance with MIL-C-53072.

3.13.5 Exceptions. Parts which can be flexed should not be painted with CARC since the paint will crack.

3.13.6 Marking for CARC. The equipment shall be marked with 'CARC' stenciled in CARC paint in close proximity to the nameplate. A contrasting color shall be used for the marking ( e.g. black marking on a green surface). The marking shall be durable enough to withstand the full environmental conditions the surface will be exposed to during field use. Plates or embossed lettering may be used. Embossed letters need not be highlighted as long as they are legible

3.14 Marking. Marking shall conform to MIL-STD-13231. Front panel marking shall be Group I as described in MIL-STD-13231.

3.14.1 Identification Label. An identification label shall be provided for each power supply. The label shall include the following information:

Power Supply
Nomenclature
NSN
Contract Number
Serial Number
Manufacturer
Manufacturer CAGE CODE

The label shall be located on a visible side of the power supply. The material used for the label shall be weather-resistant and capable of withstanding the tactical and environmental requirements of this specification.

3.15 Visibility. Whenever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with minimum disassembly of the equipment.

3.16 System Safety Requirement.

3.16.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. Verification that compliance with this requirement has been achieved shall be through a visual inspection. (see 4.8.1)

3.16.2 Edge Rounding Exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch ( 1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

3.16.3 Radioactive Materials. Radioactive materials shall not be used ( e.g. luminous dials/markings, electron tubes, surge arrestors and lenses).

3.16.4 Bleeder network. The bleeder network shall discharge the AC line filter so that the voltage at the input connector pins shall drop to 30 volts or less within two seconds after the power is turned off. Similarly, all internal components accessible to maintainers shall drop to 30 Volts within 2 seconds after power is turned off.



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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection(see 4.2),
b. Inspection covered by subsidiary documents(see 4.3),
c. Conformance inspection (see 4.4).

4.2 First Article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

4.2.1 First article units. Unless otherwise specified, the contractor shall furnish eight (8) first article units of the complete Power Supplies. The presence of one or more defects shall be cause for rejection of First Article.

4.2.2 First article inspection. The first article inspection shall consist of inspections specified in Table III. Reliability inspection shall be performed per the contractor-written, government approved plan.

4.3 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

| Item | Requirement Paragraph. |
|---|---|
| Parts and materials | 3.2 |
| Printed wiring boards | 3.5 |
| CARC | 3.13 |
| Marking | 3.14 |

TABLE III. First Article Inspection.

| Inspection | Reqt Para | Test Para | UNIT | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1. Inspections covered by subsidiary documents | | 4.3 | Inspections to be performed on all units. | | | | | | | |
| 2. Group A Inspection | See Table IV | | Inspections to be performed on all units. | | | | | | | |
| 3. Group B Inspection | See Table VI | | Inspections to be performed on all units. | | | | | | | |
| 4. Frequency-voltage-load matrix | 3.6.2 | 4.5.1.2 | Inspections to be performed on all units. | | | | | | | |
| 5. Temperature Test<br>Temperature Coefficient<br>Relative Humidity | 3.7.1<br>3.6.9<br>3.7.2 | 4.6.1<br>4.5.10<br>4.6.2 | X<br>X<br>X | | | | X<br>X | | | |
| 6. Altitude (Elevation)<br>Stability<br>Electromagnetic Interference | 3.7.3<br>3.6.8<br>3.8 | 4.6.3<br>4.5.9<br>4.9 | | X<br>X | | | | X | | |
| 7. Immersion<br>Blowing sand and dust | 3.7.4<br>3.7.7 | 4.6.4<br>4.6.8 | | | X | | | | X | |
| 8. Vibration<br>Shock, Ballistic | 3.7.5<br>3.7.5 | 4.6.5<br>4.6.6 | | | | X<br>X | | | | |
| 9. Fungus* | 3.7.6 | 4.6.7 | | | | | | | | X |
| 10. Reliability | 3.11 | 4.10 | Per approved reliability test plan. | | | | | | | |

*Note: The FAT sample undergoing Fungus inspection shall be disposed of properly.



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4.4 Conformance inspection. The contractor shall perform the inspection specified in 4.3, 4.5, and 4.4.1 through 4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to assure compliance with all specification requirements.

4.4.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections in Table IV. Power Supplies which have passed Group A inspection shall be formed into lots.

4.4.1 1 Order of inspection within Group A. Group A inspection shall be performed in any order satisfactory to the Government.

TABLE IV. Group A inspection.

| Inspection | Requirement Paragraph | Inspection Paragraph |
|---|---|---|
| Visual and mechanical | 3.12 | 4.8 |
| Input voltage(AC) | 3.6.1 | 4.5.1 |
| Output voltage | 3.6.2 | 4.5.1 |
| Power changeover | 3.6.5.3 | 4.5.5 |
| Ripple voltage | 3.6.3 | 4.5.2 |
| Efficiency | 3.6.6 | 4.5.8 |
| Air-seal | 3.10 | 4.6.9 |
| Battery discharge | 3.6.5.5 | 4.5.6 |
| Protection, general | 3.6.5.1 | 4.5.11 |
| Remote sensing | 3.6.7 | 4.5.7 |
| Overload protection | 3.6.5.2 | 4.5.4 |
| Current limiting | 3.6.5.4 | 4.5.4 |
| System Safety | 3.16 | 4.8.1 |

4.4.2 Group B inspections. Group B inspections shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. Sampling of lots shall be in accordance with Table V below. No failures shall be allowed. Tests shall be conducted in the order shown in Table VI in accordance with the inspections listed in Table VI.

4.4.2.1 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE V. Group B sampling.

| Lot Size | Samples Required 1/ |
|---|---|
| Up to 13 | ALL |
| 14 to 500 | 13 |
| 501 to 1200 | 20 |
| Over 1200 | 32 |

NOTES:
1/ No failures allowed.

TABLE VI Group B inspection.

| Inspection | Requirement Paragraph | Inspection Paragraph |
|---|---|---|
| Recovery time | 3.6.4 | 4.5.3 |
| Interchangeability | 3.9 | 4.7 |



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4.4.3 Group C inspection. Group C inspection shall consist of the inspections identified in Table VII below. Samples shall be selected in accordance with 4.4.3.1.1 or 4.4.3.1.2, as applicable.

TABLE VII Group C inspection.

| Inspection | Requirement Paragraph | Inspection Paragraph | Sample Numbers |
|---|---|---|---|
| Sub-Group 1: | | | 1-4 |
| Frequency – voltage- load matrix | 3.6.1 | 4.5.1.2 | 1-4 |
| Stability | 3.6.8 | 4.5.9 | 1-4 |
| Temperature Coefficient | 3.6.9 | 4.5.10 | 1 |
| Immersion | 3.7.4 | 4.6.4 | 2 |
| Vibration | 3.7.5 | 4.6.5 | 3 |
| Shock, Ballistic | 3.7.5 | 4.6.6 | 4 |
| Sub-Group 2 | | | 5-8 |
| Altitude (Elevation) | 3.7.3 | 4.6.3 | 5 |
| Relative Humidity | 3.7.2 | 4.6.2 | 5 |
| Temperature Test | 3.7.1 | 4.6.1 | 5 |
| Electromagnetic Compatibility | 3.8 | 4.9 | 6 |
| Fungus | 3.7.6 | 4.6.7 | 7 |
| Blowing sand and dust | 3.7.7 | 4.6.8 | 8 |

4.4.3.1 Group C sampling.

4.4.3.1.1 Sampling for inspection ( Group C, Sub-group 1). Four samples shall be selected at random, from every inspection lot and numbered 1 through 4. These units shall have passed Group A and B inspections. Each sample shall be subjected to each inspection specified for the specific sample number.

4.4.3.1.2 Sampling for inspection (Group C, Sub-group 2). This inspection shall be required only when the first article samples are not specified by contract. Four samples shall be selected at random from the first inspection lot and numbered 5 through 8. Samples may be drawn from production at any time prior to the start of Group A inspection to accommodate test scheduling. Each sample shall be subjected to each inspection specified for the specific sample number.

4.4.3.2 Group C failures. Actions relative to Group C failures shall be as specified in the contract.

4.4.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are re-subjected to and pass Group A inspection and the frequency-voltage-load matrix inspection of 4.5.1.2 (see Sub-group I, Table VII). Printed wiring assemblies shall not be repaired (3.5.1).

4.4.4 Group D inspection. This inspection shall consist of the test specified in Table VIII and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

TABLE VIII. Group D inspection.

| Inspection | Requirement Paragraph | Inspection Paragraph |
|---|---|---|
| Reliability | 3.11 | 4.10 |



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4.4.4.1 Group D failures. Failure of any Power Supply to meet the requirements of Table VIII shall be cause for rejection of that assembly.

4.4.4.2 Re-inspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are re-subjected to and passed Group A inspection.

4.5 Electrical tests.

4.5.1 Voltage tests, input and output.

4.5.1.1 Input and Output. The output voltage shall meet the requirements of 3.6.2 over the load current range of 0 to 25 amperes and over the input voltage range 103.5 to 126.5 volts/ 207 to 253 volts, 60 Hz.

4.5.1.2 Frequency-voltage-load matrix test. The output voltage shall meet the requirements of 3.6.2 when the load, input voltage, and frequency are varied as specified below; also when the external control is adjusted over the output voltage range:

| Input Voltage | Frequency | Load (amps) |
|---|---|---|
| 103.5 | 47.5, 63, 420 | 0 to 25 |
| 115 | 47.5, 63, 420 | 0 to 25 |
| 126.5 | 47.5, 63, 420 | 0 to 25 |
| 207 | 47.5, 63, 420 | 0 to 25 |
| 230 | 47.5, 63, 420 | 0 to 25 |
| 253 | 47.5, 63, 420 | 0 to 25 |

4.5.2 Ripple voltage. The peak to peak ripple voltage shall be measured with a calibrated oscilloscope having a bandwidth of 20 Hz to 10 MHz . This test shall be made using the following input voltages: 103.5 Vac- 60 Hz, 115 Vac- 60 Hz, and 126.5 Vac – 60 Hz. (see 3.6.3).

4.5.3 Recovery time. With the power supply operating from 115 Vac – 60 Hz, the load shall be switched from 25 to 0 amperes and the power supply switched on and off at each of the load conditions. The voltage transients shall be noted on a calibrated oscilloscope. This procedure shall be repeated when the load is switched from 0 to 25 amperes. The transient voltage shall meet the requirements of 3.6.4.

4.5.4 Overload protection and current limiting. With the power supply operating at full load (25 amps) the load shall be increased gradually to a short circuit condition and the output current noted during the current limiting condition. Upon removal of the short, the power supply shall automatically return to normal operating conditions. Repeat five (5) times. The power supply shall then meet the requirements of 3.6.1 , 3.6.2, and 3.6.3. This test shall be performed at room temperature and during the temperature tests of 4.6.1. The input voltage during the tests shall be 115 V rms, 60 Hz. Visual examination shall be made for any physical damage that might have occurred due to the overload conditions. (see 3.6.5.2 and 3.6.5.4).

4.5.5 Power changeover. Using a 115 V, 60 Hz single phase variable supply, reduce the AC input voltage until the power supply automatically changes over to DC and note the AC voltage at this point. Increase the AC input voltage until the power supply automatically changes back to AC and note the value of AC input voltages at this point. The two voltages shall meet the requirements of 3.6.5.3. Repeat this test using 115 V – 400 Hz and 230 V – 60 Hz and 230 V – 400 Hz input. This test shall be conducted with both a 2 amp and a 25 amp load for each input voltage and frequency specified. Proper “light” indication shall be checked for conformance to requirements of 3.6.5.3. Under the power changeover condition, indicated when the indicator light on the front panel labeled “DC EMER” is on, a check shall be performed to determine the continuity and consistency of polarity of the circuit connecting the “battery in” plug and the “power out” plug.



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4.5.6 Battery discharge. With the input power source removed from the power supply, a fully charged 24 volt secondary type battery shall be connected across the dc output in correct polarity. The battery shall then be connected in reverse polarity. The reverse current flow from the battery for either polarity shall not exceed ±100 milliamperes(see 3.6.5.5).

4.5.7 Remote sensing. With the power supply adjusted for remote sensing, the load shall be connected through a 25 foot cable, 4 conductor #12 wire size ( 2 conductors for sensing). The output voltage shall be measured at the load and shall meet the output voltage requirements of 3.6.2 ( except the voltage range may be limited to 24 Vdc to 29 Vdc) when tested as specified in 4.5.1. During this test there shall be no evidence of unstable operation(see 3.6.7).

4.5.8 Efficiency. The overall efficiency of the power supply shall be measured under conditions of full load, 28.0 volts at 25 amperes, and at an input of 115 volts rms, and at frequencies of 50, 60 and 400 Hz. The percent efficiency shall be determined by the following and shall meet the requirements of 3.6.6:

$$\frac{\text{Output power (volts x current x 100)}}{\text{Input power (measured with wattmeter)}} = \%\ \text{efficiency}$$

4.5.9 Stability. The power supply shall be operated from 115 volt, 60 Hz source with a 25 ampere load at nominal output voltage for 24 hours at room temperature. At the end of this test, the power supply shall meet the requirements of 3.6.8. Repeat this test using an input of 115 Vac, 400 Hz.

4.5.10 Temperature coefficient. The power supply shall operate from 115 volt, 60 Hz source with a 25 ampere load at nominal output for 4 hours in an ambient temperature of 70°F (21.1°C). The ambient temperature shall then be increased to 120°F (49°C) and the equipment operated for 4 hours. At the end of this test, the change in the output voltage shall not have exceeded 1.0%. Repeat this test at an ambient temperature starting at -40°F (-40°C) and increasing to +10°F (-12.2°C). ( see 3.6.9).

4.5.11 General protection(see 3.6.5.1). With the AC input adjusted to 115 Vac, 60 Hz, adjust the DC output to 28 Vdc with a 25 ampere load. Quickly raise the input voltage to 139 Vac and then reduce it back to 115 Vac. At the end of the test, the power supply shall conform to the requirements of 3.6.2, 3.6.3, and 3.6.5.4.

4.6 Environmental conditions(see 3.7).

4.6.1 Temperature test.

4.6.1.1 Low temperature. The equipment shall be tested in accordance with Method 502.4, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -80°F (-62.2°C) and the operating temperature (Step 5) shall be -40°F (-40°C). Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.7.1.1.

4.6.1.2 High temperature. The equipment shall be tested in accordance with Method 501.4, Procedure I of MIL-STD-810. Constant temperature storage shall apply (Step 3). The storage temperature shall be +160°F (+71°C). Upon completion of Step 3, an operational test shall be conducted. Adjust the internal chamber temperature to +150°F (66°C) and maintain until temperature stabilization of the test item is reached. Conduct an operational checkout of the test item; document results. Proceed with Step 4. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.7.1.2.

4.6.2 Humidity. The Power Supply shall be subjected to the humidity test specified in Method 507.4, of MIL-STD-810. Conduct operational checkout during Step 3 of each cycle and the last 5-hour period of the fifth 48-hour cycle in Step 5. Document results. Inability of the equipment to meet the requirements of 3.7.2 shall constitute failure of this test.



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4.6.3 Low pressure (Altitude). The equipment shall be tested in accordance with Method 500.4, Procedure I, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.7.3.

4.6.4 Immersion. The equipment, as prepared for field transportation, shall be tested in accordance with Method 512.4, Procedure I, of MIL-STD-810. Inability of the equipment to meet the requirements of 3.7.4 shall constitute failure of this test.

4.6.5 Vibration. The equipment shall be tested in accordance with Method 514.5, Procedure I, Category 4 (Restrained Cargo) of MIL-STD-810 using Figure 514.5C-2 of Annex A for two wheeled trailers and wheeled vehicles. Exposure duration for the vibration levels shall be ½ hour or less. The equipment shall be vibrated for ½ hour at the most severe vibration level. Inability of the equipment to meet the requirements of 3.7.5 shall constitute failure of this test.

4.6.6 Ballistic Shock(see 3.7.5). The equipment shall be tested on the "Light Weight Shock Machine" in accordance with Method 522, Procedure III, of MIL-STD-810. The equipment shall be secured in its normal operating position to the test plate. The test shall consist of a total of 3 blows: one each 5 foot blow, on the back, side and top of the test plate with the blows being directed to the equipment respectively. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. Inability of the equipment to meet the requirements of 3.7.5 shall constitute failure of this test.

4.6.7 Fungus. The power supply, with the cover removed, shall be subjected to the fungus test specified in Method 508.5 of MIL-STD-810. There shall be abundant growth colonization (see 6.5.2) on 50 percent or more of the area of the control item after 14 and 28 days. The test shall be conducted for a minimum of 28 days. No cleaning of the power supply is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.7.6 shall constitute failure of this test.

4.6.8 Sand and Dust test. The equipment shall be tested in accordance with Method 510.4, Procedure I (Blowing Dust) and Procedure II (Blowing Sand), of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.7.7.

4.6.9 Air-seal test. The equipment shall be subjected to a vacuum of 1 pound per square inch ( 1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the case. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1 minute period, the decrease in vacuum shall not exceed 0.1 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.1 pound can be determined readily(see 3.10).

4.7 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a defect.

a. External and internal dimensions of cases, covers, and insert-able subassemblies and replaceable parts, when such dimensions affect mating of parts.
b. Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.
c. Location of fasteners on separate parts or assemblies which must mate together such as cases, covers, and mountings.

4.8 Visual and mechanical inspection. The equipment shall be inspected for the defects listed in paragraphs 3.12 and 3.12.1.

4.8.1 System Safety Engineering. Visual and mechanical inspection shall be performed to verify the requirements of paragraphs 3.16.1, 3.16.2, 3.16.3, and 3.16.4.



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4.9 Electromagnetic interference (EMI). The equipment shall be tested for compliance with the requirements of 3.8 of this specification.

4.10 Reliability testing. Reliability testing shall be conducted in accordance with the contractor-written, government approved reliability test plan. If First Article is required by the contract, the reliability test plan shall be included in the First Article Test Plan. In the event First Article is not required by the contract, a separate Reliability test Plan shall be required. The contractor's plan shall address, but not necessarily be limited to, the issues of: (1) selection of the reliability test model; (2) sample selection and quantity; (3) test durations; (4) minimum acceptance criteria along with the rationale; and both the (5) consumer and (6) producer risks associated with the plan. The Conformance Group D reliability tests shall be performed on the first inspection lot, only if not performed during the First Article testing. Testing shall be described using the reliability test condition described below.

4.10.1 Reliability test condition. The reliability test shall be based on a 90% probability of providing 5,000 hours of failure-free operation. A failure for reliability purposes shall be defined as the inability of any equipment to meet the Group A inspection requirements, plus the recovery time inspection requirements of paragraph 4.5.3 in this specification. Actual test conditions shall be as specified in the contractor written, government approved test plan. The following may constitute some or all of the conditions used:

a. Any AC input condition, or combination thereof, identified in paragraph 3.6.1, or the frequency-voltage load test matrix of paragraph 4.5.1.2.
b. The full load output requirements as defined by paragraph 3.6.2.
c. Intermittent operation with a minimum of 4 consecutive hours of operating time during any 12 hour period.
d. The High Temperature requirements of 3.7.1 and/or the ambient temperature identified in paragraph 3.11 (122°F).
e. The vibration conditions of paragraph 4.6.5.

4.11 Printed wiring boards/printed wiring assemblies. All printed wiring boards and printed wiring assemblies shall be tested for compliance with the requirements of 3.5.2 of this specification.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use The Power Supply is intended to provide dc power for operation of Receiver-Transmitter RT-246( )/VRC-12 or RT-524( )/VRC-12 from a 115/230 volt, 50, 60, or 400 Hz single phase ac power source.



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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Packaging requirements (See 5.1).

c. First Article inspection:

(1) Eight sample units of each item cited in Section 1 are generally required. Lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on another sample unit. ( see 3.1)

d. Marking and shipping of samples.

e. Place of final inspection.

6.3 Group C and D failures. Approval to ship may be withheld, at the discretion of the Government, pending decision from the contracting officer on the adequacy of corrective action. ( see 4.4.3.2 and 4.4.4.1)

6.4 Location of air-seal test. It is desirable that the air-seal test (4.6.9) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Definitions.

6.6.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hyphae).

6.5.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.5.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.5.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.5.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5.6 Inspection lot. An inspection lot is any group of power supplies presented to the Government for acceptance at one time.

6.5.7 Night Vision Imaging System (NVIS) Compatibility. NVIS use image intensifier tubes to produce an enhanced image of a scene in light conditions too low for normal navigation.



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6.6 Subject term (key word) listing.

Power Supply
Battery Standby
Single Phase

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:

Army – CR

Preparing activity:

Army – CR

(Project 6130-2007-01)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at http://assist.daps.dla.mil.



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